

1 Summary

Model (Run-name): model1 (test)

Iteration: 38

TAC: 154894.53386423766

Took: 7.8901166915893555s

Total Time: 92.71894478797913s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

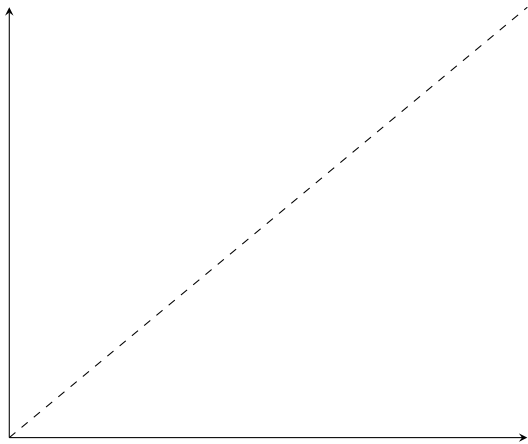
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

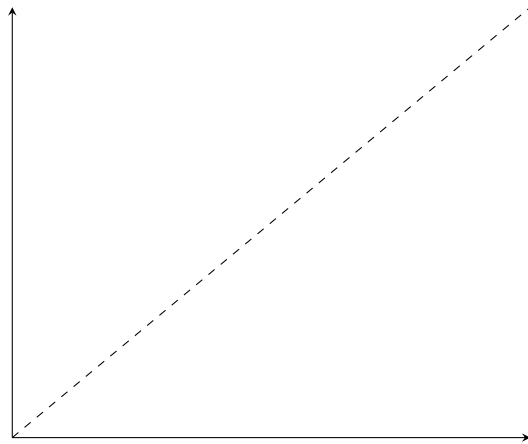
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052047083183007096	$b_{1,1,1}^{H,out}$	906.0053599430839
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017871929689700526	$b_{1,2,2}^{H,out}$	3412.2583456954753
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.0286193864154095236	$b_{2,1,2}^{H,out}$	4263.442996935106
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.0146615139220648736	$b_{1,1,1}^{C,in}$	8571.561483307765
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.0095654167403291076	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.0225122025261820446	$b_{2,1,2}^{C,in}$	5870.88041568653
$A_{1,1,1}^\beta$	71.30011482813455	$q_{1,1,1}$	685.62567779476776	$b_{1,1,1}^{C,out}$	9257.187161102533
$A_{1,2,2}^\beta$	69.70052578983292	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	138.55705332155773	$q_{2,1,2}$	2421.5614833078386	$b_{2,1,2}^{C,out}$	8292.441898994368
$A_{cu,1}^\beta$	4.8073435862836025	$q_{cu,1}$	164.37432220522533		
$A_{cu,2}^\beta$	37.84304537613103	$q_{cu,2}$	1978.4385166921566		
$A_{hu,1}^\beta$	13.307344891293862	$q_{hu,1}$	492.8128388973986		
$A_{1,1,1}$	71.30011482813455	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.70052578983292	$t_{1,2}^{(H)}$	581.4374322205224		
$A_{2,1,2}$	138.55705332155773	$t_{1,3}^{(H)}$	386.4374322205225		
$A_{cu,1}$	4.8073435862836025	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.84304537613103	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.307344891293862	$t_{2,3}^{(H)}$	468.9219258346078		
$dt_{1,1,1}$	32.854189259826576	$t_{1,1}^{(C)}$	617.1458107401734		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.4374322205227		
$dt_{1,2,2}$	81.43743222052251	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.43743222052254	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.56256777947749	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.92192583460809	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.43743222052254	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.92192583460783	$t_{1,2,2}^H$	370.0		
$dt_{hu,1}$	62.854189259826576	$t_{2,1,2}^H$	376.27968322622536		
$f_{1,1,1}^H$	2.4486631349813104	$t_{1,1,1}^C$	617.1458107401738		
$f_{1,2,2}^H$	9.222319853220034	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	11.330516068208379	$t_{2,1,2}^C$	578.7216118054197		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	1591.6310377378518		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5362.258345695483		
$f_{2,1,2}^C$	14.319220526064708	$b_{2,1,2}^{H,in}$	6685.004480242944		

3 *RecLMTD* Tangent Points

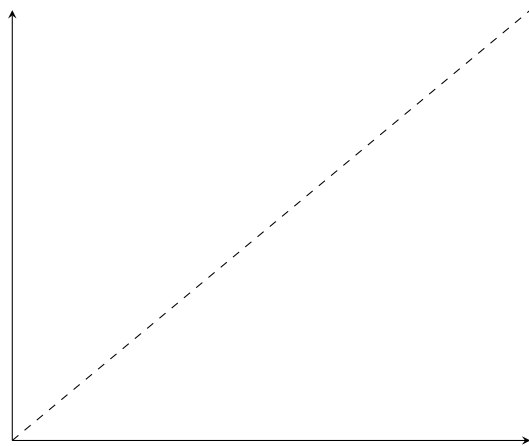
- $(1, 1, 1)$
 - Value: 0.052047
 - Error: 0.000000012306483
 - Relative error: 0.000000236448981
- $(1, 2, 2)$
 - Value: 0.017872
 - Error: 0.000000035169069
 - Relative error: 0.000001967834506
- $(2, 1, 2)$
 - Value: 0.028619
 - Error: 0.000000152257394
 - Relative error: 0.000005320050602
- $cu, 1$
 - Value: 0.014662
 - Error: 0.000000552705190
 - Relative error: 0.000037696267775
- $cu, 2$
 - Value: 0.009565
 - Error: 0.000000073849212
 - Relative error: 0.000007720378902
- $hu, 1$
 - Value: 0.022512
 - Error: 0.000000001326828
 - Relative error: 0.000000058938187



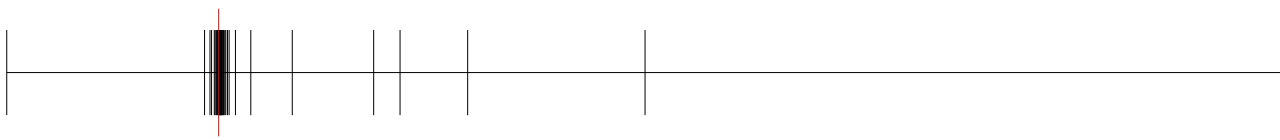
(a) *RecLMTD* - (1, 1, 1)



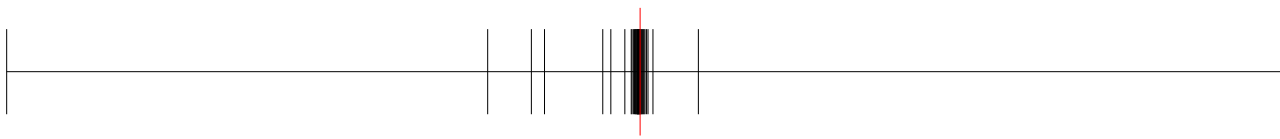
(b) *RecLMTD* - (1, 2, 2)



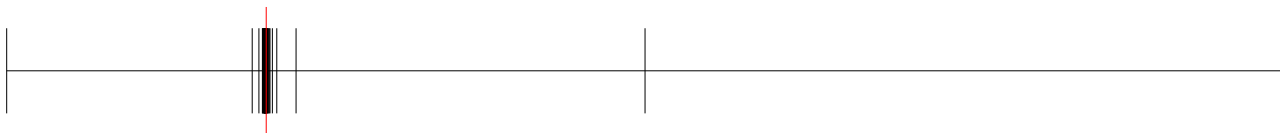
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1591.6310377378518
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5362.258345695483
 - Absolute error: 0.056371122881500924
 - Relative error: 1.0512681758130536e-05
- (2, 1, 2)
 - Value: 6685.004480242944
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 906.0053599430839
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.0038513478883421e-15
- (1, 2, 2)
 - Value: 3412.2583456954753
 - Absolute error: 4.0627128328196704e-09
 - Relative error: 1.1906228723711888e-12
- (2, 1, 2)
 - Value: 4263.442996935106
 - Absolute error: 9.094947017729282e-13
 - Relative error: 2.1332399716068536e-16

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8571.561483307765
 - Absolute error: 7.639755494892597e-11
 - Relative error: -8.912909870354623e-15
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 5870.88041568653
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9257.187161102533
 - Absolute error: 7.457856554538012e-11
 - Relative error: -8.056287968201476e-15
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 8292.441898994368
 - Absolute error: 5.599516352951468
 - Relative error: 0.000675711699872664



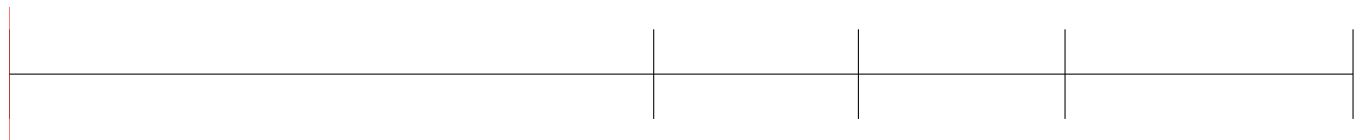
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



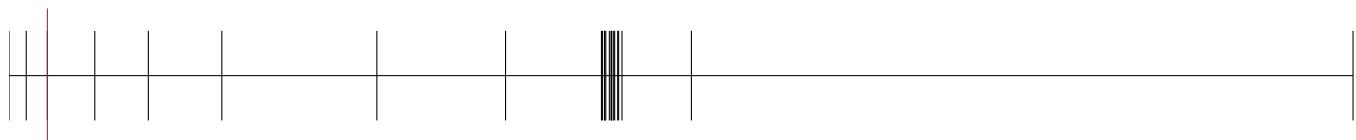
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



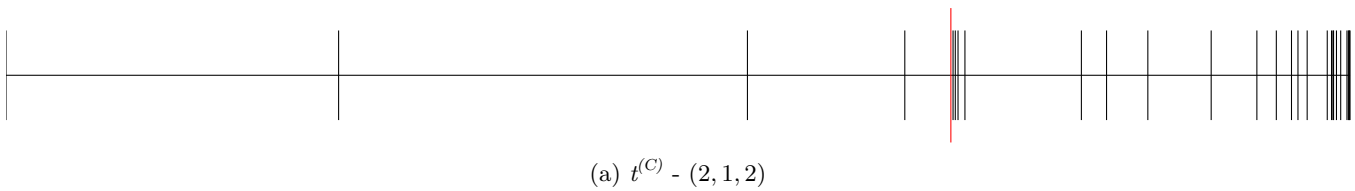
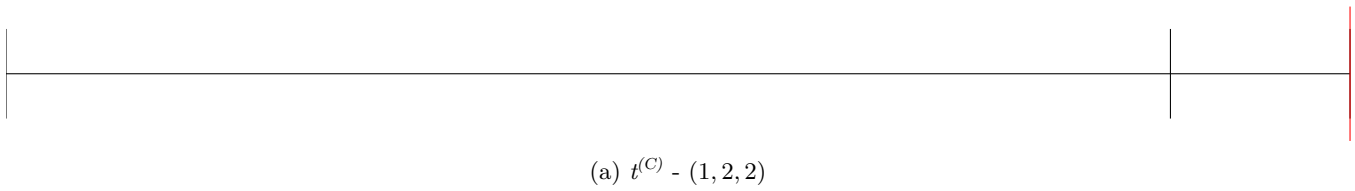
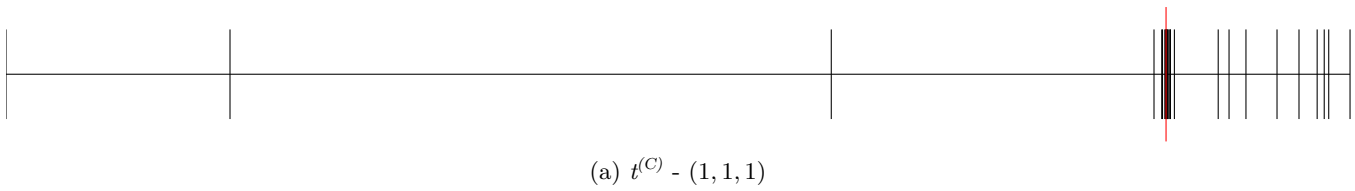
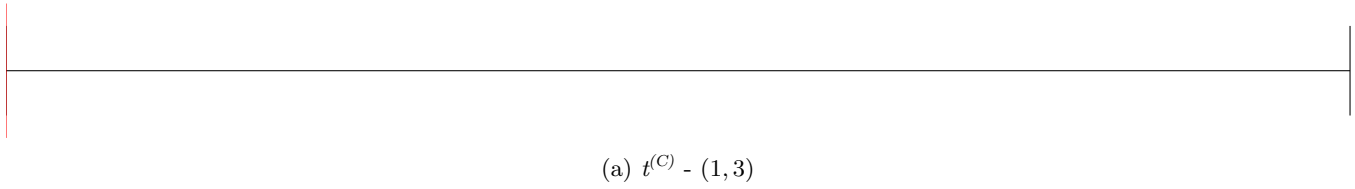
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

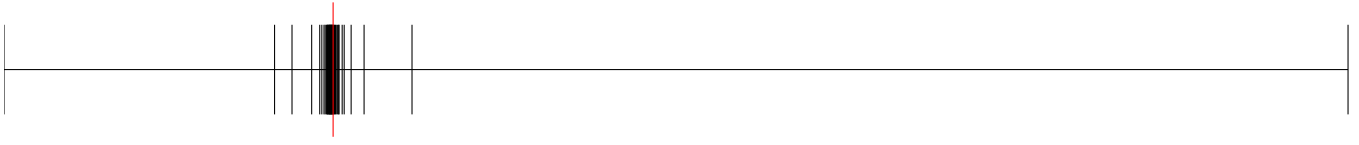
- $(1, 1, 1)$
 - Value: 71.300115
 - Error: 0.069518541045227
 - Relative error: 0.000974063306247
- $(1, 2, 2)$
 - Value: 69.700526
 - Error: 0.0000000000000895
 - Relative error: -0.000000000000013
- $(2, 1, 2)$
 - Value: 138.557053
 - Error: 0.050154317360807
 - Relative error: 0.000361844944539

5.2 A_{CU}

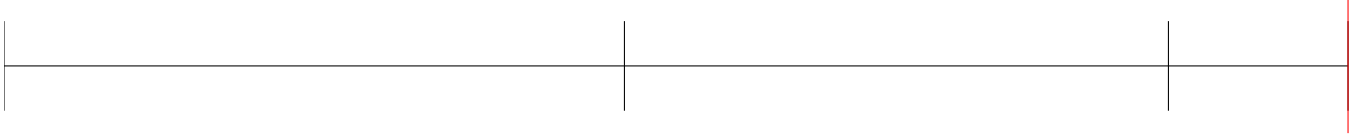
- $cu, 1$
 - Value: 4.807344
 - Error: 0.012609240600174
 - Relative error: 0.002616050623949
- $cu, 2$
 - Value: 37.843045
 - Error: 0.006132438427052
 - Relative error: 0.000162023028799

5.3 A_{HU}

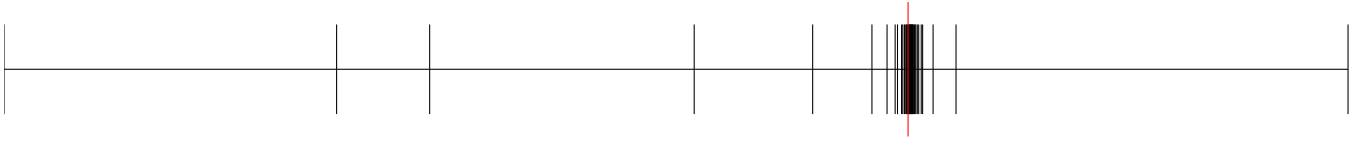
- $hu, 1$
 - Value: 13.307345
 - Error: 0.005818032819290
 - Relative error: 0.000437013567133



(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

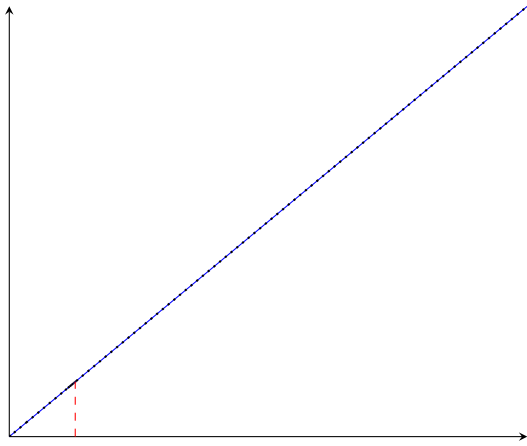


(a) $A^\beta - hu, 1$

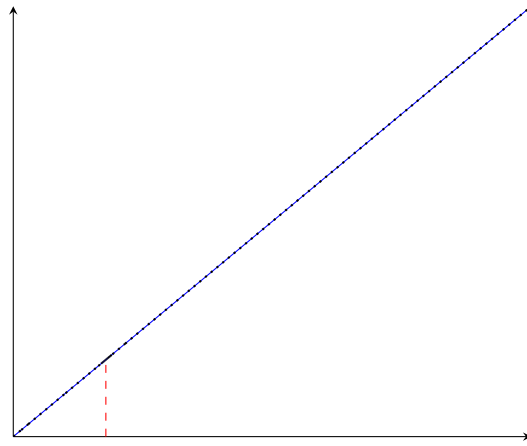
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.300115
 - $\hat{q}^\beta$: 71.300115
 - q^β : 71.300115
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.49758876855047, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, 70.65581083979713, 70.76867866839005, 70.83181816751056, 70.8571173980136, 70.94547904612048, 70.95870595008942, 71.00935653192329, 71.03465240716494, 71.1226084153824, 71.18527282143235, 71.21162487917152, 71.22933662920877, **71.2616536557244**, **71.33750386361461**, 71.3743618177527, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.700526
 - $\hat{q}^\beta$: 69.700526
 - q^β : 69.700526
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 66.74637189255778, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, 68.48329505760876, 68.57240847861726, 68.66676327863097, 68.76047282726, 68.85281320938876, 68.94418284540035, 69.0314457108091, 69.13316279873226, 69.22655548799992, 69.3178258940278, 69.34502840170047, 69.41355730879985, 69.51025044804472, **69.60484346150977**, **69.93657924014246**, 70.08419437796795, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 138.557053
 - $\hat{q}^\beta$: 138.557053
 - q^β : 138.557053
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 137.02923097052596, **137.63821724432228**, **138.86359281280824**, 139.27702937706263, 139.47623233772285, 140.0019654200325, 140.31685261310218, 140.7008934199032, 140.73764915547463, 141.47857504067568, 141.79783607898793, 142.2268253113294, 142.4338700260237, 142.98255202451594, 143.30274961189664, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 150.84969099235698, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.807344

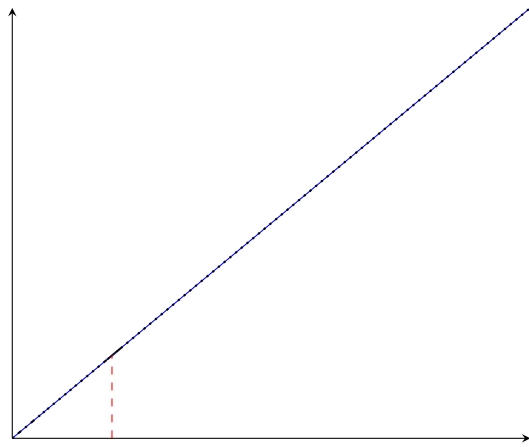
- $\hat{q}^\beta$: 4.807344
- q^β : 4.807344
- Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.707108999269609, **4.75409580038526**, **4.813467676447643**, 4.825601156120121, 4.844843256177314, 4.849719301830286, 4.882089573558703, 4.888121952381175, 4.9192700692457025, 4.956260392645724, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925, 4.993351789275153, 5.030173826900646, 5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.0892956597199035, 5.195781532076625, 5.3386847474643435, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.843045
 - $\hat{q}^\beta$: 37.843045
 - q^β : 37.843045
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.40128299925594, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.68654427850372, 37.711317605988725, 37.71812813120871, 37.738762662315665, 37.7631578546152, 37.76995088018704, 37.78573239590169, 37.791030068986316, 37.81486687126635, **37.821651654621526**, **37.87287854392874**, 37.88148462263016, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.307345
 - $\hat{q}^\beta$: 13.307345
 - q^β : 13.307345
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.064750414488246, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.224123753786348, 13.22717920580809, 13.23798359102618, 13.238585544475612, 13.244985069124171, 13.251789114535772, 13.265751604070672, 13.266498143734903, 13.269097009439808, 13.279580305153486, 13.293477718720517, **13.294574825457861**, **13.30871858348944**, 13.321736298873567, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



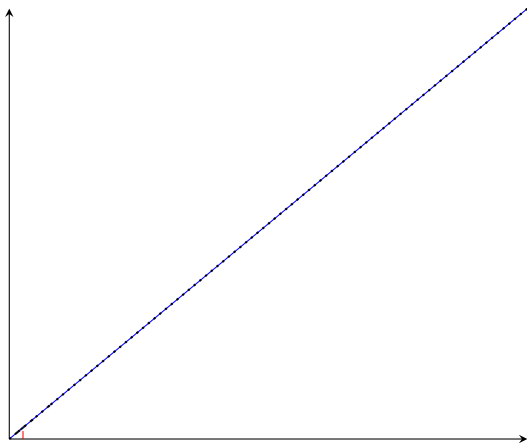
(a) $q^\beta - (1, 1, 1)$



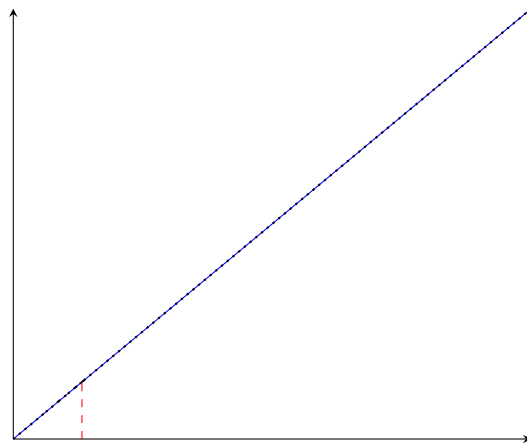
(b) $q^\beta - (1, 2, 2)$



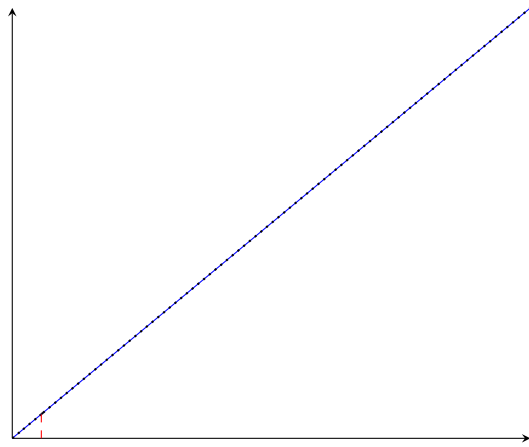
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	1346.0	$M_{2,2,2}^{H,out}$	5114.995519757051
$z_{1,2,1}$	0.0	$f_{2,2,2}^C$	0.0	$M_{1,1,2}^{C,in}$	279.1195843134698
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.013817625349338655	$M_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.01607047381606835	$M_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.0	$M_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	0.0	$RecLMTD_{2,2,1}$	0.0045138888888888625	$M_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.004340277777777778	$M_{1,1,2}^{C,out}$	279.11958431347375
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$M_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$M_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$M_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$M_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	568.7597413746703		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	589.8231946089388		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	220.0	$t_{2,1,1}^C$	564.1579740786403		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	350.0		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	452.1159765097609		
$f_{1,1,2}^H$	0.7776801467799663	$b_{1,2,1}^{H,in}$	4908.368962262148		
$f_{1,2,1}^H$	7.55133686501869	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	5114.995519757051		
$f_{2,2,2}^H$	8.669483931791621	$b_{1,1,2}^{H,out}$	452.1159765097527		
$f_{1,1,2}^C$	0.6807794739352921	$b_{1,2,1}^{H,out}$	4908.36896226214		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		